

Build explainable tuning experiments.

A complete guide to installing DroneDream, creating a PX4 tuning study, reviewing simulation evidence, and preserving the boundaries that keep every engineering decision auditable.

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Engineering boundary: independently reproduce selected parameters in SITL and inspect the complete logs before considering real hardware.

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01 Start here

DroneDream is a local-first workspace for configuring, simulating, and comparing PX4 controller parameters. Language models clarify intent and prepare reviewable drafts; deterministic validation, constraints, acceptance rules, and human review keep every experiment reproducible.

- Use Windows 10 or Windows 11 on an x64 computer.
- Reserve at least 52 GiB on a writable NTFS drive for the isolated runtime.
- Create an account before saving user-scoped settings or cloud records.
- Configure a model only for Tuning Chat or an LLM-guided strategy.
- Treat every result as simulation evidence, never automatic hardware approval.

02 Install and prepare

Download version 1.0.0 from the official website and install the desktop application on an eligible local drive. On first launch, DroneDream prepares a dedicated WSL2 distribution for PX4, Gazebo, workers, and experiment artifacts without reusing a personal Ubuntu distribution.

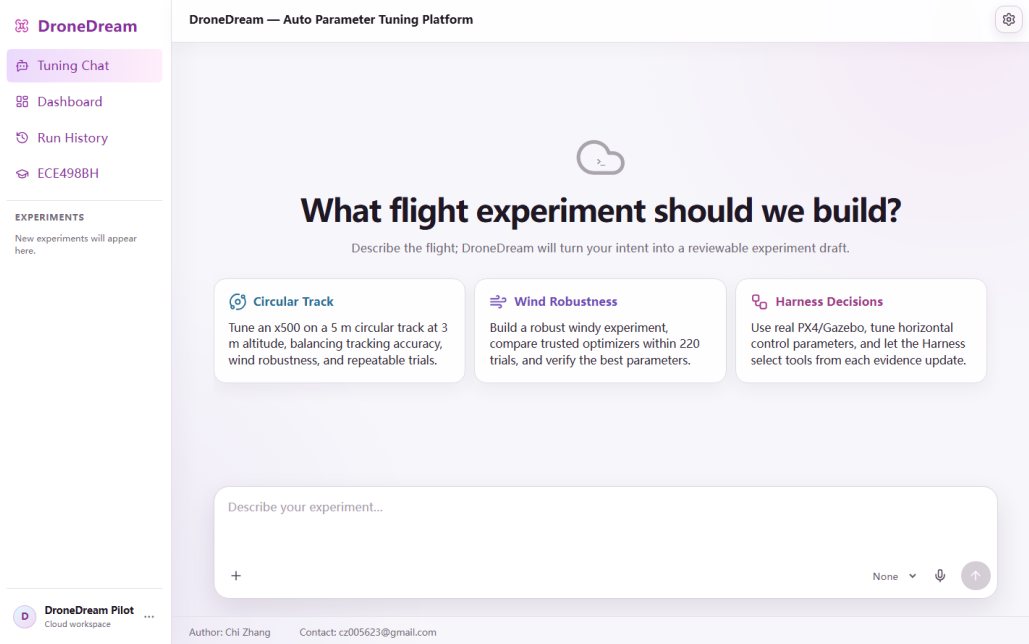
01	Download	Get the current Windows installer from getdronedream.com .
02	Install	Choose an eligible NTFS drive and keep the recommended directory.
03	Prepare	Wait while the isolated runtime is imported and verified.
04	Enter	Open the workspace only after the green status reads Checked.

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Opening the workspace does not start another check. Run a new check only from Settings > Local runtime > Check environment.

03 Create with Tuning Chat

Describe the aircraft, route, target behavior, disturbances, and priorities in plain language or by voice. DroneDream extracts explicit values, lists missing decisions, and prepares a draft that must still be reviewed before any job can begin.



Example: Tune an x500 quadrotor on a 5 m circular track at 3 m altitude. Prioritize tracking accuracy, include moderate sensor noise, and keep the experiment within 180 trials.

- The assistant summarizes the intended study in engineering terms.
- Known fields are placed into the draft with visible provenance.
- Missing parameter ranges and acceptance thresholds remain open for confirmation.
- The assistant cannot create a running job or bypass the five-step review.

04 Complete the five-step experiment

The manual workflow exposes every field and records the current stage. Reopening a draft returns to the same experiment and the last completed step, so detailed configuration can be reviewed without rebuilding the study.

01	Flight Setup	Choose the vehicle, world, objective weights, and track.
02	Parameters	Confirm each selected PX4 parameter and its search range.
03	Scenarios	Define search, holdout, wind, noise, seeds, and vehicle effects.
04	Constraints & Budget	Select a strategy and set trial and acceptance limits.
05	Review	Audit the complete contract before creating the experiment.

05 Edit a custom flight track

The track editor pairs a plot with an equal-height coordinate table. Switch among XY, XZ, YZ, and 3D views, edit full X/Y/Z values, and use JSON import or export when a path is easier to generate with another tool.

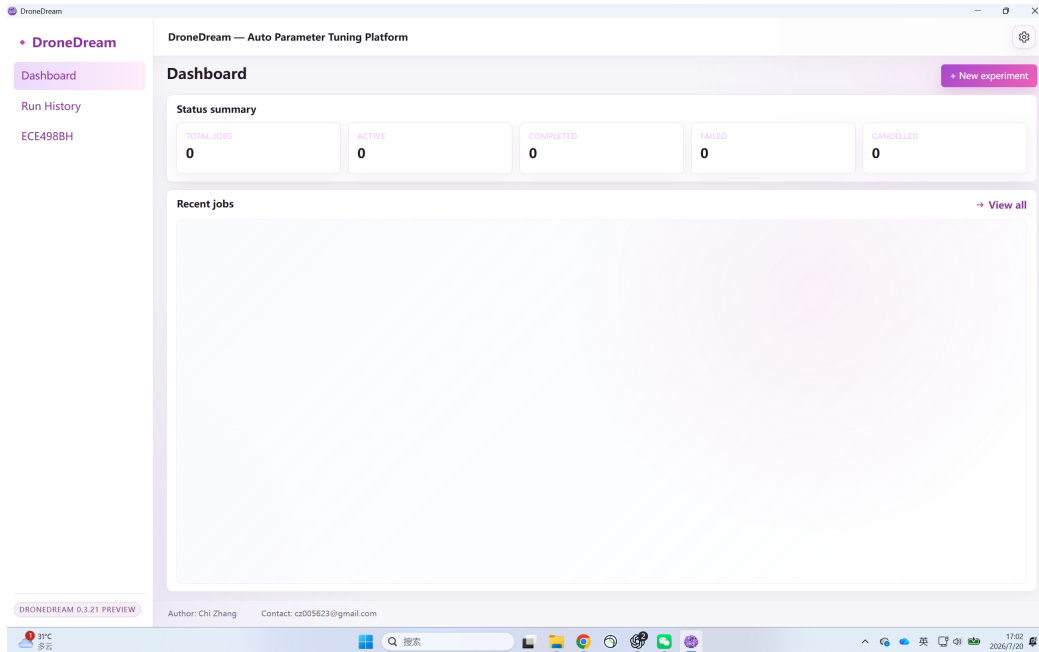
- Select a row to highlight the corresponding waypoint.
- Add, undo, or delete points with the controls above the coordinate table.
- Keep imported coordinates inside the intended simulation volume.
- In 3D, both ground axes use the same real-world unit and square grid cells.

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Extending a route adds more square cells; it must never stretch the existing ground grid into rectangular units.

06 Review history and evidence

Dashboard and Run History expose completed, failed, cancelled, and active experiments. Filter by status, track, objective, strategy, or date, then open a result to inspect the metrics, scenarios, logs, artifacts, and configuration that produced the decision.



01	Configuration	Vehicle, firmware, route, ranges, constraints, and budget.
02	Execution	Scenario identity, seeds, manifests, logs, and artifacts.
03	Decision	Feasibility, error, overshoot, settling, robustness, and trade-offs.

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Compare only experiments whose scenario and metric contracts are compatible. A lower score is not meaningful when validation conditions differ.

07 Accounts, data, and safety

Supabase identity and row-level policies isolate cloud account records. Public community topics are deliberately shared, while account settings and user-owned cloud data remain scoped to their owner. Local drafts stay available during the application session and are discarded after a confirmed exit.

- Provider and model settings may be saved on the device, while the model API key remains only in session memory.
- API keys are never written into experiment drafts, run records, exports, or community posts.
- DroneDream assists engineering judgment; independent SITL validation and operator approval remain mandatory before hardware flight.